💥 CANNON DUEL: TRAJECTORY BATTLE TICKET

2D PROJECTILE KINEMATICS • ANGLE • VELOCITY • ELEVATION COMPENSATION

DAY 45 IN-CLASS TICKET

Pilot Name: _____ **Rival / Co-Pilot:** _____ **Period:** _____ **Date:** _____

🎯 TACTICAL SIMULATION INTEL (HTTPS://RRMUDRY.GITHUB.IO/CANNON_DUEL/)

1. Distance & Elevation (📐 Measure): Click the 📐 tool to read the exact horizontal span (Δx) and height difference (Δy) between forts.

2. Ideal Gravity (g = 10 m/s²): World physics has zero wind and zero air drag: flight time is governed entirely by vertical motion.

3. Reconnaissance Feedback: The impact flag reports your exact error in meters (e.g. *"12 m SHORT"* or *"8 m LONG"*).

PART 1: 5-SHOT TELEMETRY & BRACKETING LOG (RECORD 5 DISTINCT TURNS) — STANDARD CANNONBALL OR HEAVY SHELL

SHOT #	TARGET DISTANCE & HEIGHT (READ WITH 📐 MEASURE)	LAUNCH ANGLE Θ (DEG)	LAUNCH SPEED V_0 (M/S)	RECON IMPACT RESULT (METERS SHORT / LONG / HIT)	TACTICAL ADJUSTMENT FOR NEXT TURN
Turn 1	Δx = _____ m , Δy = _____ m	_____ °	_____ m/s	_____	☐ Angle ↑/↓ ☐ Speed ↑/↓
Turn 2	Δx = _____ m , Δy = _____ m	_____ °	_____ m/s	_____	☐ Angle ↑/↓ ☐ Speed ↑/↓
Turn 3	Δx = _____ m , Δy = _____ m	_____ °	_____ m/s	_____	☐ Angle ↑/↓ ☐ Speed ↑/↓
Turn 4	Δx = _____ m , Δy = _____ m	_____ °	_____ m/s	_____	☐ Angle ↑/↓ ☐ Speed ↑/↓
Turn 5	Δx = _____ m , Δy = _____ m	_____ °	_____ m/s	💥 DIRECT HIT / Fort Blast!	Target Knockout ✓

PART 2: 2D KINEMATICS STRATEGY & PHYSICS ANALYSIS — THINK • APPLY • JUSTIFY

Q1. Correcting a "SHORT" Landing: If your cannonball lands 25 meters SHORT of the enemy fort, state <u>two different physics adjustments</u> you could make to reach the target on your next turn:

Q2. High Arc vs. Direct Fire: Both 35° and 55° can hit a target across flat ground at the same speed. Which angle stays in the air longer? Why does a higher launch angle increase hang-time?

Q3. Firing Uphill (Enemy on High Cliff, +Δy): When the opposing cannon sits on a mountain high above your fort, why must you increase your launch speed compared to a flat shot?

Q4. Heavy Shell vs. Standard Cannonball: In the ammo rack, the Heavy Shell creates a massive blast. Why does its trajectory follow the *exact same parabolic arc* as a standard light ball at the same angle & speed?

🤝 Rival / Co-Pilot Verification:

"I verify my partner actively aimed, fired 5+ rounds, and recorded authentic telemetry data."

Rival Signature: _____

MR. MUDRY STAMP / SIGN-OFF

Score: _____ / 10 pts